AlphaStock in Indian Stock Market
From Paper Understanding to Full Implementation

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Roadmap

1. Transition from Proposal to Execution
2. Data Pipeline for Indian Market
3. Model and Training
4. Results and Diagnostics
5. Takeaways

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- Pipeline commitment: Bloomberg data $\rightarrow$ cleaning $\rightarrow$ feature engineering $\rightarrow$ RL model.
- Extension idea: replace paper's LSTM encoder with simpler MLP to reduce overfitting risk.
- Target metric remained Sharpe ratio (risk-adjusted performance), not only raw returns.

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- Full implementation notebook: `alphastock_mlp.ipynb`.
- Training, evaluation, and diagnostics plots exported for reporting.

Data Engineering Workflow

Bloomberg Export $\rightarrow$ Raw CSVs $\rightarrow$ Semi-cleaned Data
$\rightarrow$ Final Clean Data $\rightarrow$ Normalized Features $\rightarrow$ RL States/Rewards

- Universe aligned to Indian equities (NSE/BSE tickers present in cleaned datasets).

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- Monthly feature creation and rolling normalization to avoid leakage.
- Reward definition kept consistent with long-short AlphaStock formulation.

Feature Engineering (Implemented)

```
def get_normalised_data(file):
    df = pd.read_csv("final_clean_data/"+file, index_col=0)
    df['ret_1m'] = np.log(df['price'] / df['price'].shift(1))
    df['ret_3m'] = np.log(df['price'] / df['price'].shift(3))
    df['mom_12m'] = df['price'] / df['price'].shift(12) - 1
    df['vol_3m'] = df['ret_1m'].rolling(3).std()
    df['vol_6m'] = df['ret_1m'].rolling(6).std()
    # ... volume, valuation, size and composite factors ...
    rolling_mean = df[features].rolling(24).mean().shift(1)
    rolling_std  = df[features].rolling(24).std().shift(1)
    df[features] = (df[features] - rolling_mean) / (rolling_std + 1e-8)
    df['R'] = (df['price'].shift(-1) / df['price'])
```

State and Reward Interface

```
def get_state_reward(time):
    month = months[time]
    state_i, reward_i = [], []
    for file in files_list:
        try:
            stock_data = pd.read_csv('normalised_data/'+file, index_col=0)
            s_i = stock_data.loc[month][features]
            r_i = stock_data.loc[month]['R']
            state_i.append(s_i)
            reward_i.append(r_i)
        except:
            continue
    return state_i, reward_i
```

Architecture Extension: MLP + Cross-Stock Attention

```
class MLPEncoder(nn.Module):
    def __init__(self, input_dim, hidden_dim, output_dim, dropout=0.2):
        super().__init__()
        self.net = nn.Sequential(
            nn.Linear(input_dim, hidden_dim), nn.BatchNorm1d(hidden_dim),
            nn.ReLU(), nn.Dropout(dropout),
            nn.Linear(hidden_dim, hidden_dim), nn.BatchNorm1d(hidden_dim),
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- Original AlphaStock idea preserved: learn stock representations then model cross-asset dependencies.
- Practical change: LSTM replaced with compact MLP encoder.

Cross-Asset Attention (Core AlphaStock Mechanism)

```
class CrossStockAttention(nn.Module):
    def forward(self, r):
        Q = self.WQ(r); K = self.WK(r); V = self.WV(r)
        scores = torch.matmul(Q, K.T) / math.sqrt(self.d_model)
        alpha = F.softmax(scores, dim=1)
        A = torch.matmul(alpha, V)
        s = self.ws(A).squeeze(-1)
        return s
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- Winner score for each stock is conditioned on all other stocks.
- Enables relative ranking needed for long-short portfolio construction.

Sharpe-Oriented Loss

```
def compute_loss(R_list, H0=0.0):
    R = torch.stack(R_list)
    mu = R.mean()
    sigma = R.std() + 1e-6
    sharpe = mu / sigma
    cumulative = torch.cumprod(1 + R, dim=0)
    peak = torch.cummax(cumulative, dim=0).values
    drawdown = ((peak - cumulative) / (peak + 1e-8)).max()
    loss = -0.5 * (sharpe - H0) - 0.5 * mu * 12 + 0.2 * drawdown
    return loss
```

- Maintains paper objective while adding practical drawdown regularization.

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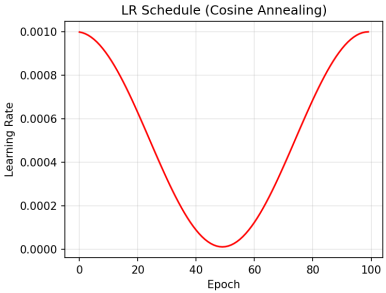
Observed Training Behavior

Epoch Sharpe moved from ~0.48 to ~1.78 with stable learning-rate schedule.

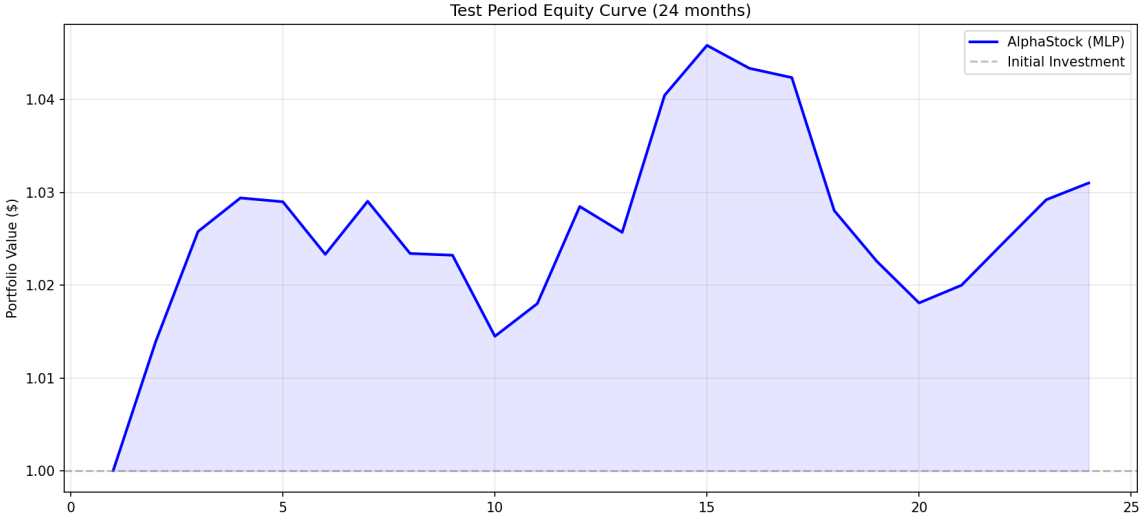
Training Curves



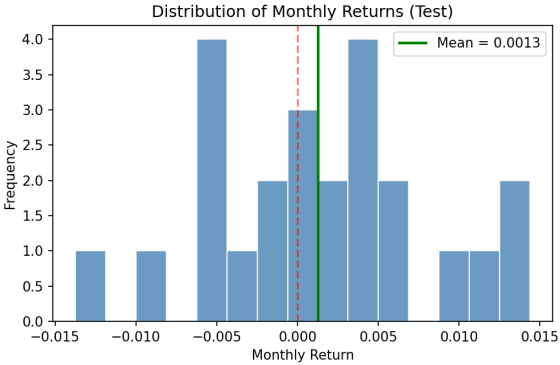
Test Performance (24-Month Holdout)

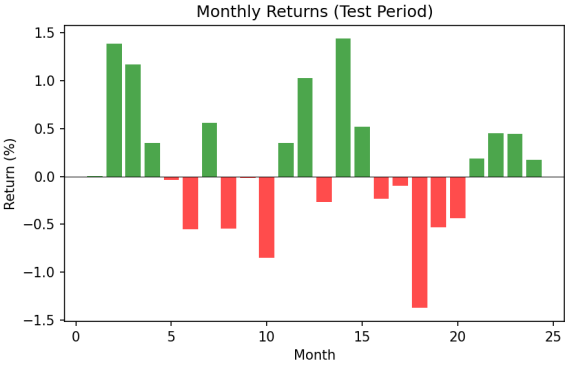
Metric	Value
Monthly Sharpe	0.1863
Annualized Sharpe	0.6454
Total Return	3.11%
Final Equity ($1 base)	1.0310
Annualized Return	1.54%
Max Drawdown	2.65%

Equity Curve and Drawdown

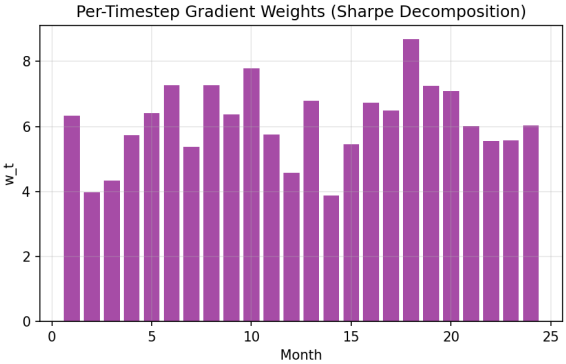


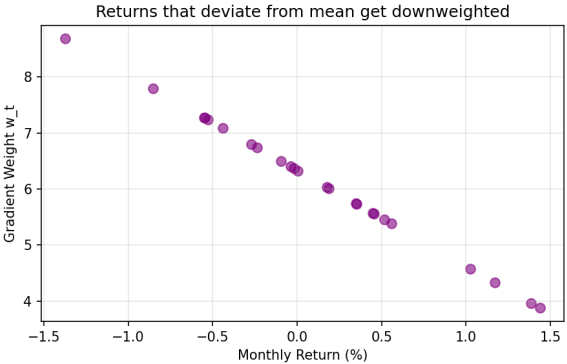
Return Distribution





Sharpe Gradient Weight Diagnostics





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- Practical model variant (MLP encoder) can preserve core AlphaStock logic with lower parameter complexity.
- Risk-aware training objective yields controlled drawdown with positive holdout performance.
- Codebase is reusable for future extensions: actor-critic, transaction costs, regime-aware features.

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- Expand to multi-horizon training windows and stress periods.

Thank You